

# Agent Memory Selection under Bounded Context: State, Provenance, and Trajectory Economics

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## Abstract

Long-running coding agents accumulate instructions, actions, observations, edits, and verification state. We study which typed logical records can be omitted independently of the inference engine. Decomposing each record into state, provenance, control, and semantic evidence exposes a central negative result: operationally superseded state is not necessarily behaviorally irrelevant. Removing it can erase localization or workflow evidence and trigger reacquisition or failure.

We evaluate boundary-free causal policies with mini-swe-agent on SWE-bench Verified using restored workspaces, exact-prefix controls, repeated full-history qualification, and official resolution as the primary endpoint. A Recent Frontier policy preserves 3/3 qualified discovery successes, uses 22 versus 30 calls, and omits 40.53% of its own full-history counterfactual. A Prompt Pinned policy separately preserves 3/3 cold identities, uses 33 versus 34 calls, and saves 39.45% within trajectory and 41.71% against paired Full. However, in two frozen eight-identity orders, Full resolves only 3/8 and 1/8. Recent Frontier preserves the three eligible first-order successes at 40.05% paired saving, but the second order contributes only a 0%-saving abstention. These cohorts qualify mechanisms, not population accuracy.

A post-refactor audit also finds that historical H2/T4 manifests recorded floors the materializer did not consume. Corrected mini-swe-agent H2/T4 preserves 3/3 solves but saves only 6.03% as whole causal groups and adds three calls. Corrected exact-token OpenHands transfer preserves 3/3 solves, uses 74 versus 77 actions, and saves 22.67% against its own-trajectory counterfactual and 29.76% against paired Full. A prospective common-model admission then separates patch quality, autonomous completion, action cost, and token work. On one locked identity, Pi and PRA Agent preserve the Full solve while reducing calls and candidate-own input; OpenCode adds four calls, and Kilo requires an external stop after producing a resolving patch. These are admission tests, not cross-agent accuracy estimates. In the first prospective persistent Pi pair extended to five tasks, M2/P1 preserves every Full success (3/3), loses none, and adds one recovery diagnostic. It resolves 4/5 rather than 3/5, saves 40.92% against its own trajectory and 37.51% against paired Full, and uses 155 versus 152 calls. Charging two discarded transport-failed Task-5 attempts reduces paired saving to 29.05% and changes the call delta from +3 to +27. The completed six-task selective trajectory resolves 5/6 and saves 51.36% against its own counterfactual, but has no valid paired N=6 denominator because Full's cold 53,484-token Task-6 request produces no response before the frozen timeout.

A stricter decoding audit finds that nominal temperature zero is not an exact trajectory control on the served Qwen endpoint: identical requests vary from one valid action to twelve code blocks while the model retains a top- $k$  of 20. Explicit top- $k = 1$  makes the selective first request identical in 5/5 repeats. Under that control, a task-neutral static workflow anchor plus closed-task P0/W0 retirement preserves the official Task 7 solve and cuts paired cumulative input by 75.24%, but uses 14 calls rather than Full's six. After replacing fragile shell editing with a matched, digest-recorded transactional capability and correcting its resource receipt, Full and P0/W0 again both resolve. The selective arm uses 30,647 tokens versus Full's 213,298, an 85.63% paired reduction, but uses eight versus six calls and seven versus six executed actions. This stronger mechanism point preserves capability and removes edit/recovery churn, while still

failing the strict no-call-increase gate on one identity. It also does not transfer as a prompt mechanism: on Task 4, moving a stricter anchor from the system prefix to the current user instruction yields a repeatable frozen next action but autonomous FULL still reaches 25 calls without verification or submission and never invokes the installed transactional command.

Finally, per-request omission can reverse end-to-end: one six-issue policy resolves 6/6 and omits 55.34% against its own trajectory, yet expands from 60 to 114 calls and consumes 43.87% more input than paired Full. We therefore report own omission, paired cumulative work, task success, and action cost separately. The surviving policies are promotion candidates rather than production defaults; native K/V realization belongs to a separate runtime contract.

## 1 Introduction

Software-engineering agents differ from ordinary retrieval-augmented prompts in one crucial way: their context is not a static evidence set. It is a growing history of actions and consequences. A coding trajectory may contain the original task, source-code observations, edits, failed commands, test results, intermediate hypotheses, and final verification. Many of these records become irrelevant quickly, while others remain necessary long after they were created.

The usual history policy is recency. Given a context limit, an agent keeps the latest messages or tokens and discards older state. This is simple but ignores causal structure. An action and its observation form a unit. A source view may remain important after several later path-discovery commands. A mutation can remain relevant until a subsequent verification. A test result may be obsolete after another edit, while the task statement remains permanently relevant.

This paper isolates the memory-policy problem from the serving-engine problem. At model turn  $t$ , let the full canonical history be

$$H_t = [r_1, r_2, \dots, r_n],$$

where each  $r_i$  is a logical agent record. A memory policy computes

$$S_t = \pi(H_t, Q_t, B),$$

where  $Q_t$  is the active task/state and  $B$  is a context budget. The selected plan  $S_t$  is then serialized to an ordinary model request. No native K/V mechanism is required for the main experiments. This serialization is an experimental consumer, not proof that an already-resident K/V cache can realize every transformation without new model evaluation.

This separation is deliberate. Prior PRA work studies retrieval, native-memory materialization, cross-document composition, and engine-specific K/V reuse. Those systems questions should not determine which semantic records an agent needs. Here we first identify useful logical memory policies. A separate runtime paper can later ask whether different engines realize the same selected plan correctly and efficiently.

We begin with mini-swe-agent because its control loop is minimal: a linear message history is sent to the model, the model emits shell actions, the environment executes them, and observations are appended to the same history. This minimizes hidden harness state and makes memory intervention easy to audit.

**Research questions.** We ask:

1. How much agent history can be omitted while preserving official task resolution on paired, repeat-qualified executions?
2. Which parts of historical state remain behaviorally useful after operational supersession, and what does that imply for record granularity?

3. When logical context shrinks, does total trajectory work also shrink after reacquisition, extra calls, and submission behavior are counted?

**Contributions.** This paper contributes:

1. An engine-independent typed record model that separates state, provenance, control, and semantic-evidence utility.
2. A fail-closed evaluation contract combining oracle counterfactuals, matched recency, exact-prefix controls, and repeat-qualified autonomous SWE-bench runs.
3. Evidence that operational supersession does not imply behavioral irrelevance, making whole-record semantic retirement an unsafe default.
4. Two promotion-candidate boundary-free policies and trajectory-level efficiency endpoints that charge reacquisition and additional calls.

## 2 Problem Formulation

At decision  $t$ , the canonical logical history is

$$H_t = [r_1, \dots, r_n], \quad S_t = \pi(H_t, Q_t, B),$$

where  $Q_t$  is the live query and workspace state,  $B$  is a logical budget, and  $S_t$  is the selected plan. The key unit is not an undifferentiated token window. We represent a record as

$$r = (s, p, c, e),$$

where  $s$  is authoritative world or resource state,  $p$  is provenance,  $c$  is workflow/control state, and  $e$  is semantic evidence that can remain useful to reasoning. These components can expire at different times. In particular,

$$\boxed{\text{state superseded} \not\Rightarrow \text{memory useless.}}$$

A newer file view can supersede older bytes as state while the earlier search, localization, failed attempt, or verification remains useful provenance or control evidence. This distinction motivates causal bundles and payload-level retirement rather than blind deletion of complete turns.

We keep four accounting objects separate:

**Logical history.** Every canonical agent record produced so far.

**Selected history.** Existing records or spans admitted by  $\pi$ .

**Realized trajectory.** All calls, actions, reacquisition, recovery, and completion tokens caused by the treatment.

**Engine realization.** The physical text or K/V mechanism used to consume the frozen logical plan; this is outside the primary policy experiment.

## 2.1 Causal units and component lifetimes

The record decomposition is operational rather than merely descriptive. A tool observation can stop being the authoritative copy of a file while its resource identity still establishes provenance, its return code still closes a control branch, and a diagnostic line still supplies semantic evidence. A selector therefore assigns retention at component or span granularity and keeps the causal identity of the parent record. It must not silently turn an assistant command and its observation into unrelated messages.

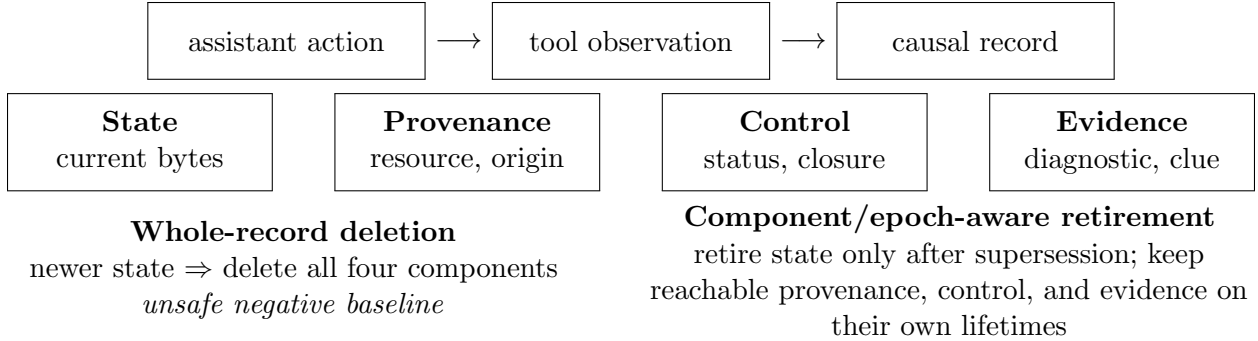


Figure 1: A causal record is not one disposable semantic unit. Whole-record deletion conflates state authority with future trajectory utility; structured retirement preserves parent identity while assigning component-specific lifetimes.

Short records remain intact. Oversized observations are subdivided only at natural boundaries—files, diff hunks, test cases, stack frames, or lines—and each child keeps a stable parent and causal-group identifier. The current incomplete turn, genuine user prompts, recent mutation state, and submission or verification state are control floors rather than ordinary retrieval candidates. These rules make the plan independent of mini-swe-agent’s textual format even though Bash command semantics provide the first concrete tool ontology.

## 3 Evaluation Contract

The evaluation uses mini-swe-agent’s ordinary Bash interaction on SWE-bench Verified with Qwen3-Coder-30B, temperature zero, restored repository snapshots, and hash-bound history plans. Each treatment is admitted only after a successful full-history execution from the identical saved prefix. Cold-cache qualification is used when endpoint state changes behavior; repeated full-history controls measure residual trajectory instability. A timeout- or transport-contaminated execution is quarantined rather than counted.

For the frozen confirmation campaign, “cold” is executable rather than a label: before every Full and candidate trial the runner sends Ollama `keep_alive=0`, performs the same temperature-zero OpenAI-path warmup, then verifies the active model and 131,072-token context. The unload, warmup, latencies, and context check are persisted in the trial receipt; failure pauses the cell before a benchmark action is taken.

The outcome hierarchy is explicit. *Solution-state success* means that the terminal workspace contains a resolving patch. *Submission success* means that the agent emitted the required valid benchmark payload. *Official resolution* is the primary endpoint. Auxiliary workspace grades diagnose reasoning versus protocol failure but never replace official resolution. No treatment is credited when its paired full-history qualifier fails.

### 3.1 Paired execution protocol

Every autonomous comparison begins from one immutable prefix and workspace snapshot. Full first determines whether that model/task/host cell is eligible; the treatment then consumes the same prefix with only the memory plan changed. The evaluator preserves both trajectories rather than retrying a legitimate submission failure away. Infrastructure faults are quarantined with a reason code and never pooled with model or policy outcomes.

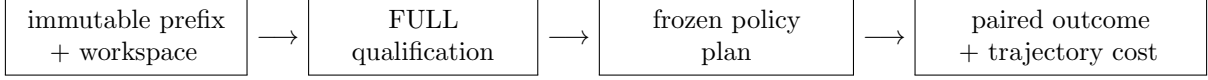


Figure 2: Fail-closed paired protocol. A failed Full cell cannot be used to claim either policy failure or saving.

The statistical unit is the task identity, not an individual execution row. Repeated runs estimate within-task instability; they do not manufacture extra independent samples. Cohorts with different tasks, prefixes, models, or cache conditions are reported separately. This prevents overlapping campaigns from being pooled into an apparently larger accuracy estimate.

Cross-agent and cross-engine comparisons use different controlled axes. A cross-agent result keeps the provider, exact model and tokenizer revisions, chat template, task snapshots, policy parameters, and generation settings fixed; only native-event normalization, declared tool semantics, and protocol-valid serialization may vary. A cross-engine result instead keeps the agent fixed and replays one exported selection ledger. History digest, plan digest, selected record identities, and logical token counts must match exactly across engines. Page rounding, resident bytes, K/V copy, temporary storage, and latency may differ, but a logical-saving difference is a failed comparison rather than an engine effect. These checks are executable in `comparison_contract.py`; the locked matrix is `configs/transverse_agent_engine_comparison_v1.json`. The strict runners also query the serving endpoint before any task action and fail closed unless its observed model digest equals the declared revision; a model alias in the request is not identity evidence. Keeping the model fixed within each axis does not by itself support a joint agent-by-engine claim. The predeclared bridge therefore pins Qwen2.5-Coder-7B-Instruct revision `c03e6d35` at BF16 for mini-swe-agent, OpenHands, and OpenCode across HF, vLLM, and MLX. The Qwen3-Coder-30B cross-agent cohort is a separate replication and is not pooled with that bridge.

The first execution of this common-model bridge has not yet crossed its plain admission gate. On the three frozen Django identities, the direct-MLX mini-swe-agent diagnostic made no workspace mutation in 18, 15, and 12 calls, respectively. A dedicated 35-call replay of the middle identity exposed a deterministic no-progress cycle among malformed or no-op edits, source grep, and unavailable interactive editors. These executions are diagnostics rather than accuracy rows: no selective policy was applied, and no engine or policy comparison is inferred from a model-agent pair that did not first solve the plain task. The corresponding request receipts and economical interaction ledger are archived by Paper 4.5.

Let  $M_t$  be selected model-visible input and let  $F_t^{\text{cand}}$  be the full-history counterfactual along the candidate’s own realized trajectory. We report

$$S_{\text{own}} = 1 - \frac{\sum_t M_t}{\sum_t F_t^{\text{cand}}}, \quad S_{\text{paired}} = 1 - \frac{\sum_t M_t^{\text{cand}}}{\sum_t M_t^{\text{Full}}}.$$

The first isolates selector omission; the second includes changed call count and trajectory. Recovery cost is

$$C_{\text{recovery}} = C_{\text{reacquire}} + C_{\text{trajectory inflation}},$$

where reacquisition counts repeated reads, searches, and tests, while trajectory inflation counts additional model calls/actions. We also report first action divergence and failure-aware saving, which assigns no efficiency credit to a treatment that loses a paired success.

## 4 Policy Families

The main comparison uses four public policy families; implementation ladders and rejected parameterizations are retained in the audit appendix.

| Policy          | Main rule                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Full            | Preserve the complete canonical history; correctness and trajectory-cost control.                                                                 |
| Matched Recency | Retain the newest records at a matched model-visible budget.                                                                                      |
| Recent Frontier | Preserve genuine prompts plus the frozen M2/P1 recent progress/protocol frontier; retire older unreachable causal groups.                         |
| Prompt Pinned   | Preserve every genuine prompt and the frozen E2+F1C instruction-epoch floors; compact closure receipts make this a transformed-context candidate. |

Table 1: Main policy families. Symbolic implementation names are secondary to the semantic distinction.

Selection-only plans retain or omit existing records and are directly transferable to resident-K/V selection. Ingest-time transforms compact a tool result before its first model evaluation. Post-cache transforms replace an already evaluated record with new receipt or summary text; they require new model evaluation and are never classified as native cache reuse.

### 4.1 Boundary-free dependency retirement

The structured policies do not receive an explicit “new task” marker. They operate over a continuous session and infer whether old causal groups remain reachable from recent prompts, resource versions, mutations, verification, and submission state. A completed group can be retired only when no retained node depends on its output and a newer authoritative resource version exists where state is involved. Absence of a dependency is used as a candidate signal, not as proof that the record’s evidence is useless.

This distinction is especially important across multiple independent issues in one developer session. File and workspace identities usually separate the issues, so old tool payloads should become unreachable as the session grows; user prompts remain pinned. For related or dependent issues the graph instead keeps cross-issue edges through shared resources, mutations, or explicit references. The policy therefore approaches high asymptotic omission only for independent completed work, without requiring the application to announce task boundaries.

### 4.2 Retention lifecycle by evidence role

The public policies differ less in their scoring function than in the evidence they refuse to discard. Table 2 states the intended lifecycle without binding it to one agent protocol. “Retire” means omit the payload from future model input while preserving stable identity and audit metadata; it never means delete the underlying workspace artifact.

| Role       | Examples                                               | Candidate retirement condition                                           | Mandatory floor                                          |
|------------|--------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------|
| State      | source view, directory listing, generated file         | newer authoritative version and no reachable dependent                   | current mutation and unresolved error state              |
| Provenance | search path, command parameters, resource origin       | resource identity remains recoverable from retained graph                | task prompts and active resource bindings                |
| Control    | hypothesis, next action, return code, submission state | branch is closed and closure is represented                              | incomplete turn, latest verification, pending submission |
| Evidence   | diagnostic, failed test, explanatory source fragment   | bounded oracle or policy evidence that removal does not trigger recovery | recent decision spine and explicit user constraints      |

Table 2: Evidence lifecycles used by structured policies. The conditions are conservative candidate rules, not proofs of behavioral irrelevance.

Recent Frontier implements these floors directly: keep recent completed causal groups, recent mutations, recent verification, and protocol state, then retire older unreachable groups. Prompt Pinned uses instruction epochs: genuine user prompts remain visible, recent epochs remain complete, and older completed epochs may retain closure evidence. The latter currently mixes selection and post-cache materialization, which is why its logical result cannot be promoted unchanged as a strict native-K/V profile.

The resulting deployment interpretation is deliberately modest. A conservative profile can use Full or a large Recent Frontier and primarily guard against unbounded session growth. A balanced profile may use the repeat-qualified Recent Frontier parameters. Prompt Pinned is an experimental high-saving profile until selection-only closure evidence and broader task/model replication pass. None of these labels overrides the paired success and workload gates.

## 5 Results

### 5.1 Operational supersession is not behavioral irrelevance

The early semantic-retirement experiments reject a tempting assumption: a read or search result does not become useless merely because a newer operation supersedes its bytes. Whole-record retirement changes the first eligible action, increases rereading/recovery, and can lose the official submission. Matched recency is not a strong final policy—across three same-host identities Full resolves 5/6 executions and Matched Recency 4/6 while saving only 6.35% of pooled treatment-history tokens—but it is an important control because it can preserve behavioral evidence that the semantic deletion rule discards. The result is

state authority and trajectory utility are different quantities.

The offline leave-one-bundle-out oracle finds 24.5–38.2% removable-history headroom on three successful trajectories, but safe removals are non-additive and non-monotone. This supports structured selection while ruling out a simple “delete everything superseded” rule.

## 5.2 Trajectory economics dominate request-level retention

Per-request saving can reverse after the model reacts to missing context. In one six-issue persistent session, aggressive instruction-epoch retirement omits 55.34% against its own trajectory and resolves 6/6 versus Full’s 5/6, but expands from 60 to 114 calls. Against the paired Full denominator it uses 43.87% more input. Other arms similarly spend their nominal saving on source reacquisition, edit recovery, or repeated verification. Therefore

logical omission  $\nrightarrow$  workload saving.

The structural opportunity does grow with session length, but it is only an upper bound. Figure 3 contrasts the equal-size opportunity with observed input saving through three accumulated issues. Extrapolated dashed and dotted curves are not accuracy results; the observed points show why persistent-session accounting is needed before native K/V experiments.

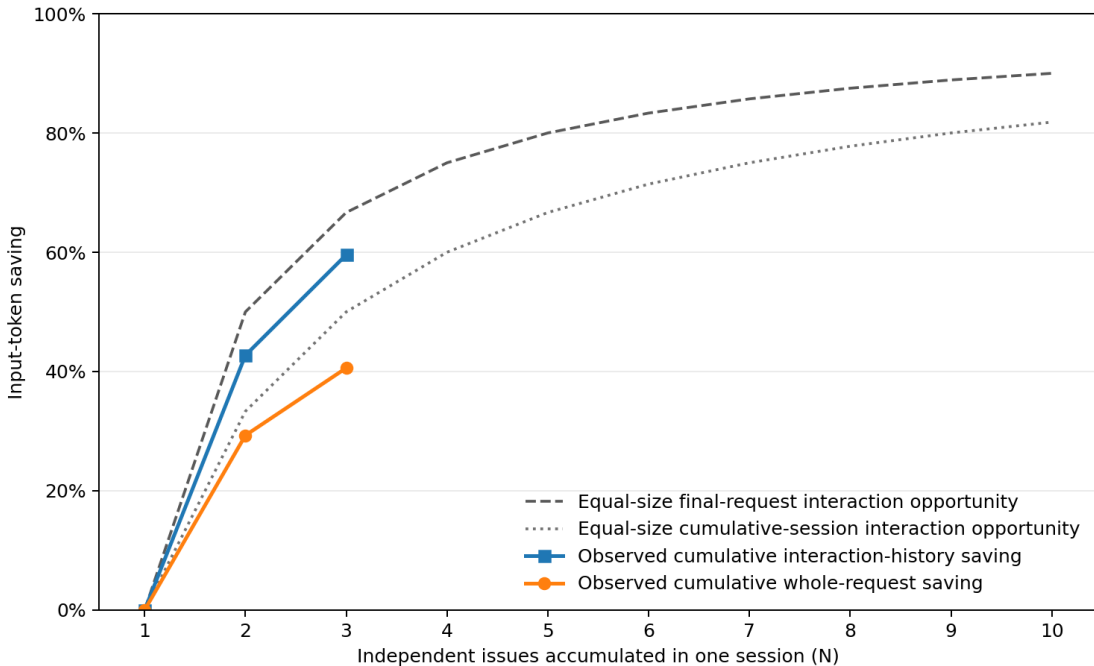


Figure 3: Instruction-epoch opportunity and observed saving as independent issues accumulate in one session. Lines beyond  $N = 3$  are structural equal-size opportunities, not autonomous quality measurements.

## 5.3 Recent Frontier discovery cohort

Recent Frontier applies recent, mutation, verification, and protocol floors over a continuous session without requiring an explicit task-boundary signal. The strict same-prefix Task 3–5 cohort is the strongest current paired point.

The cohort preserves all three qualified paired successes and reduces aggregate calls, but savings range from 11.52% to 52.56% within trajectory. It is a promotion candidate, not evidence of greater-than-90% population preservation or a universal default.

## 5.4 Prompt Pinned discovery cohort

Prompt Pinned preserves every genuine user prompt, the active instruction epoch, the two most recent completed epochs, and compact closure evidence for older completed epochs. Cold exact-prefix execution removes a cache-state confound and yields one task-clustered observation per identity.

Prompt Pinned is conceptually distinct from Recent Frontier and reaches the 30–50% target band on this small cohort. Its closure receipts are model-visible post-cache transformations in the current transfer experiments. Two direct-host selection-only E2 executions, including one cold model reload, repeat exactly and preserve 2/2 official solves on one identity at 40.56% own-trajectory saving without synthetic receipts. This qualifies strict repeatability for one task; additional identities remain necessary before recommendation as a native-PRA default.

## 5.5 Post-refactor matched-recency audit

An implementation audit found that the historical matched-tail manifests recorded two protected head and four protected tail turns although the materializer consumed neither parameter. Reducing the immutable artifacts reproduces every archived total, and four fresh Task 2–3 FULL controls reproduce their old official outcomes, calls, and cumulative tokens exactly. The refactor therefore did not alter the baseline or reducer; the historical treatment is a valid prompt-pinned pure-recency policy with a superseded H2/T4 label.

The corrected H2/T4 arm keeps the first two and latest four complete causal turns. On the same three identities it preserves 3/3 official resolutions, but uses 64 calls versus 61/69 for FULL-A/FULL-B. It saves 6.03% through whole-causal-group retirement and 8.17% after record-internal span materialization against its own trajectory counterfactual, plus 6.28% against pooled FULL-A input; the task-macro paired value is  $-3.93\%$  because Task 2 expands from 15 to 22 calls. Thus corrected H2/T4 passes the observed quality gate but fails the predeclared no-call-increase efficiency gate. A second corrected Task 1 run also resolves, but uses 26 rather than 20 calls, exposing residual trajectory variance. Neither corrected nor legacy matched recency is promoted as a default.

The corrected OpenHands transfer supplies a useful contrast without changing the selector. OpenHands 1.49.2 on Tasks 1–3 resolves 3/3 officially and uses 74 rather than 77 actions. Task 1 emits the same 689-byte patch digest as its paired successful FULL control and saves 35.35% against its own logical history and 43.53% against paired FULL. Task 2 produces a different but resolving patch, saves 9.12% against its own counterfactual and 6.12% paired, and adds one action. Task 3 also resolves with a semantically equivalent patch, uses 17 versus 19 actions, and saves 8.14% own and 18.75% paired. Aggregated over task identities, H2/T4 saves 22.67% own and 29.76% paired input. The 8.14–35.35% range arises because Task 1 contains an oversized early source view that can retire once more specific reads supersede it. This three-identity result is not a population accuracy estimate, but it demonstrates that the corrected policy is transverse to the agent interface and that opportunity is workload-dependent.

A same-model bridge also bounds that claim. Qwen2.5-Coder-14B has one successful Task 1 mini-swe-agent execution, but repeated FULL controls do not make that identity a reliable admission pair; it also fails OpenHands FULL admission before any history selection. Provider-native calling first terminates on narrative tool syntax. Explicit prompt-mocked calling repairs dispatch, yet the default OpenHands surface repeats five identical invalid editor calls; a terminal-only surface instead performs broad, corruptive substitutions on Task 1 and invents an ineffective API change on Task 4. Removing unused tool schemas lowers the initial Task 1 prompt from 7,739 to 5,578 tokens but does not restore capability. We therefore do not launch a selective arm or pool this diagnostic with the qualified Qwen3-Coder-30B cohort. Agent prompts, tool protocols and tool affordances remain

part of the treatment even when model weights and decoding controls match.

The next exact transfer prevents a favorable-only interpretation. Pi 0.75.3 solves the same task under a fresh 28-call FULL control, but corrected H2/T4 fails in 28 calls. It saves 8.53% of materialized ordinary-text input while retiring only 0.30% as whole causal groups. The first changed input precedes the first changed tool action; after repeated edit failures the agent copies a partial fragment over a complete source file, producing a 91,283-byte patch that fails the official grader. The apparent saving mostly comes from rewriting one selected oversized source view, not omitting reusable records. We therefore supersede generic boundary rewriting with a declared-safe rule: an observation may be subdivided only into stable natural child spans created before model-visible materialization and K/V encoding. Otherwise the policy retires the whole causal group or leaves budget unused. This negative result is a policy-transfer failure, not an inference-engine result.

The repaired Pi reruns validate that distinction. Declared-safe H2/T4 and H2/T8 both resolve officially and reproduce the FULL control’s exact 689-byte patch digest. H2/T4 saves 45.16% against its own whole-record counterfactual but expands to 43 model calls and saves only 1.40% against paired FULL input. The predeclared H2/T8 mitigation reduces the expansion to 39 calls, saves 41.53% within trajectory and 11.81% against paired FULL, but still fails the no-call-increase and paired-economics gates. Its first 13 actions match FULL; the first divergence follows selection at request 13 and a third schema-invalid edit result. Thus the unsafe rewrite was a real correctness bug, while the remaining inefficiency is an agent recovery/conditioning result requiring causal add-back rather than a wider tail-size grid.

The first normalized Kilo pairing adds a complementary negative result. Kilo injects a wall-clock `Message time` field into the model-visible user record; a generic request-normalization declaration replaces only that volatile field before both hashing and forwarding. The resulting Full and declared-safe H2/T4 arms have the same initial request digest and generation controls, resolve officially, and emit the same 689-byte patch. Nevertheless, their first canonical tool actions differ at action 1, seven requests before selection first changes an input. The candidate saves 7.30% against its own whole-record counterfactual but materializes 376,980 selected tokens versus 146,013 in Full, or  $-158.18\%$  paired saving. We therefore reject causal attribution and promotion: temperature zero and identical first inputs do not guarantee a repeatable backend trajectory.

## 5.6 Prospective cross-agent session-length admission

The v3 matrix restarts agent transfer prospectively with an exact model artifact within each paired cell, one tokenizer, one task order, and one reducer. The  $N = 1$  admissions use the 64K Qwen3-Coder-30B runtime alias; persistent  $N = 3$  uses the same model family with a separately digested 131K alias and therefore does not support byte-identical cross- $N$  comparisons. The matrix also makes official resolution and autonomous semantic completion independent outcomes: a resolving patch does not qualify an agent that hangs or is externally stopped. Table 3 reports the locked  $N = 1$  admission cell before any  $N = 3$  or  $N = 6$  expansion.

| Agent     | Full solve/calls | Tail solve/calls | Autonomous | Own save | Paired save | Decision              |
|-----------|------------------|------------------|------------|----------|-------------|-----------------------|
| OpenCode  | 1/1; 15          | 1/1; 19          | yes        | 12.48%   | 59.86%      | reject: +4 calls      |
| Pi        | 1/1; 49          | 1/1; 27          | yes        | 52.68%   | 77.03%      | admit to $N = 3$      |
| Kilo      | 1/1; 24          | 1/1; 13          | no         | 9.14%    | 82.74%      | reject: external stop |
| PRA Agent | 1/1; 19          | 1/1; 16          | yes        | 20.53%   | 23.95%      | admit to $N = 3$      |

Table 3: Prospective common-model  $N = 1$  admission on `django_django-15277`. All rows use the same 64K Qwen3-Coder-30B artifact, temperature zero, seed zero, and the same tokenizer revision. Tail-90 is a hard ceiling over complete causal groups, so realized retention can be below 90%. One identity admits a mechanism to the next gate; it is not an accuracy estimate.

This incremental cell covers agents that lacked a current declared-safe pair; the established mini-swe-agent and OpenHands exact-token cohorts remain the reference anchors. Codex is also excluded from the table rather than assigned a misleading zero. Its native Task-1 Full artifact resolves officially, but Codex 0.153 carries history through Responses `previous_response_id` state. It requires a Responses-native selector and identical-subset reference before a selective arm is comparable; translating it to Chat Completions would change the treatment.

Pi’s 52.68% own saving is not evidence that an exact 10% removal produced a large gain. One oversized early causal group cannot fit under the 90% ceiling without unsafe post-cache rewriting, so realized retention falls to roughly 29–41% on later requests. The correct remedy is to define stable child spans before their first materialization, not to compact already encoded K/V.

The first PRA-Agent  $N = 3$  Recent-Frontier M2/P1 execution produces resolving patches for all three tasks in 39 requests and omits 33.19% of its own full-history counterfactual. It remains unpaired and unpromoted. An HTTP 500 after the final verified patch makes autonomous completion false, while the fresh contemporaneous Full control fails Task 1 in a repeated inspection-budget recovery loop. A historical 3/3 Full control is retained as a stability observation but cannot supply paired economics because its trajectory differs before selection acts.

Pi now supplies the first completed prospective  $N = 3$  pair. Both arms share the exact 49-request Task 1 prefix. Full resolves the two Django tasks but exports an empty patch for Scikit-learn (2/3; 115 calls); M2/P1 resolves all three (3/3; 107 calls). The policy preserves both Full successes, so the baseline-conditional result is 2/2 with zero losses. Its 2,042,277 materialized tokens are 20.05% below its own 2,554,440-token whole-history counterfactual. Relative to Full’s different 2,847,310-token trajectory the paired saving is 28.27%, but this mixes selection with eight avoided calls. M2/P1 correctly abstains on Tasks 1–2 because the last two instruction epochs cover the complete session. On Task 3 it retires 49 Task 1 causal groups, saves 52.34% against its own trajectory, and uses 27 rather than 35 calls. Because Full fails that task after Pi’s structured-edit arguments trigger a recovery loop, the Task 3 success is a selective-recovery diagnostic, not a paired-preservation win. This single repeat admits the frozen mechanism to a repeat and  $N = 6$  gate; it is not a population accuracy estimate or a production default.

The prospective extension now reaches an exact matched five-task prefix and a completed six-task selective trajectory. Across the matched first five identities, persistent FULL resolves 3/5 in 152 calls, while M2/P1 resolves 4/5 in 155 calls. All three FULL successes are preserved, none is lost, and the Scikit-learn success remains a recovery diagnostic. M2/P1 materializes 2,857,534 tokens: 40.92% below its own 4,836,401-token counterfactual and 37.51% below FULL’s 4,572,641 tokens. This clean logical result does not hide execution failures. Two discarded Task-5 attempts add 24 calls and 386,878 materialized tokens; charging them reduces paired saving to 29.05% and yields 179 versus 152 calls. Thus the clean policy is nondominated in token work and official resolution, but the observed backend reliability does not pass the no-call-increase operational gate.

After an exact rollback-safe retry, M2/P1 completes all six episodes in 194 calls and resolves

5/6 officially. It materializes 3,446,052 of 7,085,344 whole-history counterfactual tokens, for 51.36% own saving. The sole failure, `django_django-15741`, also fails under matched persistent FULL. No N=6 paired-saving claim is made: FULL’s first Task-6 request contains 53,484 message-content tokens and produces no response before the frozen timeout. M2/P1’s first Task-6 requests retain roughly 23–24% of logical history and execute, demonstrating an operational reach benefit without supplying a matched quality denominator. These are one-order, single-repeat results and require another order/model/agent before profile promotion.

The auditable manifests and fail-closed reduction are under `docs/papers/shared/results/paper8_5_agent_memory/cross_agent_matrix_v3/`.

## 5.7 Task-clustered headline comparison

Table 4 keeps discovery and confirmation separate. Matched Recency contains six executions over three identities and is shown as a negative control; its historical schema lacks three immutable pairing digests and is not pooled with the exact-prefix cohorts. The frozen Recent-Frontier held-out confirmation now completes all eight predeclared identities under the baseline-conditional missingness rule. Prompt Pinned’s separate held-out arm remains at its first independently cold identity; its same-prefix FULL repeat fails identically and therefore launches no treatment.

| Policy                       | Cohort      | IDs                      | Resolved F/P | Calls F/P      | Input F/P    | Save own/paired       | Status                                 |                                                  |
|------------------------------|-------------|--------------------------|--------------|----------------|--------------|-----------------------|----------------------------------------|--------------------------------------------------|
| Matched Recency              | development | 3 (6 runs)               | 5/6; 4/6     | 154/113        | not admitted | 6.35%/n.a.            | two losses; incomplete legacy identity |                                                  |
| Matched H2/T4                | Recency     | post-fix audit           | 3            | 3/3; 3/3       | 61/64        | 364,457/341,556       | 6.03L/8.17M; 6.28P%                    | +3 calls; not promoted                           |
| Matched H2/T4                | Recency     | OpenHands post-fix       | 3            | 3/3; 3/3       | 77/74        | 932,038/654,626       | 22.67/29.76%                           | −3 actions; three-task transfer                  |
| Matched H2/T4                | Recency     | Pi post-fix              | 1            | 1/1; 0/1       | 28/28        | 464,135/424,805       | 0.30L/8.53M; 8.47P%                    | lost success; superseded                         |
| Declared-safe H2/T8          |             | Pi repair audit          | 1            | 1/1; 1/1       | 28/39        | 464,135/409,335       | 41.53/11.81%                           | same patch; +11 calls                            |
| Declared-safe H2/T4          |             | Kilo pairing audit       | 1            | 1/1; 1/1       | 20/19        | 146,013/376,980       | 7.30/−158.2%                           | preselection fork; noncausal                     |
| Recent M2/P1                 | Frontier    | OpenHands single epoch   | 1            | 1/1; 1/1       | 25/33        | 186,635/287,257       | 0.00/−53.91%                           | exact abstention; divergence before selection    |
| Recent M2/P1                 | Frontier    | Pi prospective $N = 3$   | 3            | 2/3; 3/3       | 115/107      | 2,847,310/2,042,277   | 20.05/28.27%                           | preserves 2/2; one recovery                      |
| Recent M2/P1                 | Frontier    | Pi matched $N = 5$       | 5            | 3/5; 4/5       | 152/155      | 4,572,641/2,857,534   | 40.92/37.51%                           | diagnostic preserves 3/3; one recovery; +3 calls |
| Recent M2/P1                 | Frontier    | Pi selective $N = 6$     | 6            | —; 5/6         | —/194        | —/3,446,052           | 51.36/n.a.%                            | Full Task 6 timeout; no paired claim             |
| Transactional anchor + P0/W0 |             | matched Task 7 mechanism | 1            | 1/1; 1/1       | 6/8          | 213,298/30,647        | 89.70/85.63%                           | +2 calls, +1 action; not promoted                |
| Recent M2/P1                 | Frontier    | development              | 3            | 3/3            | 30/22        | 556,072/239,881       | 40.53/56.86%                           | calls decrease                                   |
| Prompt E2+F1C                | Pinned      | development              | 3            | 3/3            | 34/33        | 924,686/539,022       | 39.45/41.71%                           | transformed receipts                             |
| Recent M2/P1                 | Frontier    | held-out                 | 8            | 3/8; 3/3 elig. | 37/37 elig.  | 829,884/497,538 elig. | 15.59/40.05%                           | no eligible loss; Full only 3/8                  |
| Recent M2/P1                 | Frontier    | held-out order 2         | 8            | 1/8; 1/1 elig. | 27/27 elig.  | 153,654/153,654 elig. | n.a./0.00%                             | exact abstention; Full only 1/8                  |
| Prompt E2+F1C                | Pinned      | held-out                 | 1/8          | 0/1; —         | 14/—         | 97,604/—              | —                                      | Full failed; no candidate                        |

Table 4: Task-clustered policy ledger at cutoff. F/P denotes Full/policy. Discovery and held-out confirmation are never aggregated; repeated runs expose instability but do not increase the identity denominator. For the completed Recent-Frontier row, “elig.” conditions policy accuracy and paired saving on the three identities solved by contemporaneous exact-prefix FULL; 15.59% is the unconditional whole-session saving after carrying five failed FULL episodes. The second order has one qualified eligible identity and seven failed controls; its only admitted policy arm abstains at 0% saving, so no unconditional saving is reported. The Pi  $N=5$  row is the exact matched prefix; charging its two discarded Task-5 attempts changes paired saving from 37.51% to 29.05% and calls from 155 to 179. The  $N=6$  row reports own saving only because persistent FULL does not complete Task 6.

The OpenHands single-epoch row is deliberately a scope and variance control, not a failed saving policy. With only one visible user instruction, the last  $M = 2$  instruction frontier contains the entire session, so M2/P1 has no older epoch eligible for retirement. Its 0% own saving is exact. The 53.91% paired cost increase follows a response-9 fork under identical selected message content and therefore cannot be attributed to selection. This control prevents the persistent-session policy from

being misreported as a single-task compressor and reinforces task-clustered repeat controls.

The Pi prospective row applies the same rule across three uninterrupted tasks. Its first two exact abstentions and Task 3 retirement demonstrate the expected session-length geometry: a two-epoch frontier cannot save earlier-task history until a third instruction epoch exists. The table reports candidate-own and paired savings separately; only the former isolates logical selection. Complete traces, official grader reports, hashes, and the fail-closed pair reduction are under `cross_agent_matrix_v3/pi/n3/`, with the aggregate ledger in `pair_evidence.json`. The compact  $N=5/N=6$  extension is under `cross_agent_matrix_v3/pi/n6/compact_evidence_v1/`; its summary separates clean logical economics from discarded transport attempts and binds the large immutable traces by SHA-256.

The held-out rows make the uncertainty visible rather than inflating the denominator. In the first order, exact-prefix FULL resolves only 3/8 identities; M2/P1 preserves all three eligible successes and their aggregate 37-call workload while reducing paired input by 40.05%. Carrying the five failed FULL episodes through the same persistent session reduces the unconditional saving to 15.59%. In the reversed order, FULL resolves only 1/8 and the sole eligible M2/P1 arm is an exact 0%-saving abstention. These observations qualify a mechanism on three first-order paired successes, not a population preservation rate.

Diagnostics on failed controls are likewise kept out of the preservation denominator. The normalized first order recovers one cross-task failure, but the reversed order recovers 0/6 while omitting 48.98% of candidate-own input and increasing calls from 92 to 104. Thus old-task interference sometimes matters, but recovery is prefix- and order-sensitive and does not explain the cohort generally. The companion technical report retains the task order, runtime normalization, submission analysis, and complete failure ledgers.

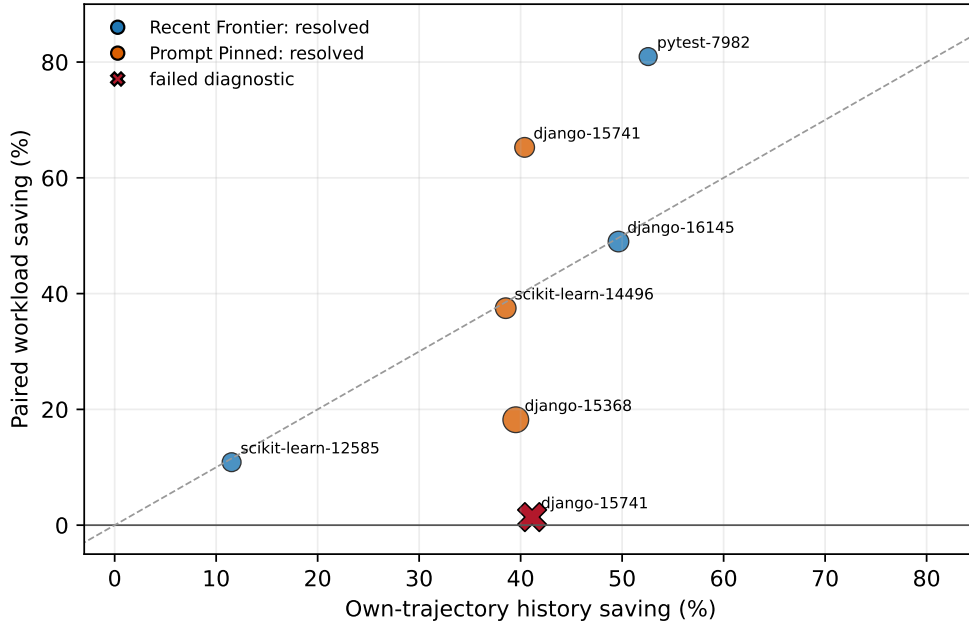


Figure 4: Own-trajectory omission versus paired workload saving. Marker area is proportional to policy calls; filled circles preserve the paired official solve and the cross marks the failed Recent-Frontier diagnostic. High omission can collapse after trajectory inflation.

## 6 Profile Implications

The evidence supports a staged candidate vocabulary, not one universally best selector. Candidates differ in how much uncertainty they tolerate and in which evidence floors they preserve. Table 5 separates frozen discovery points from controls and unqualified treatments.

| Profile              | Logical policy                    |        | Non-negotiable state                                 | Current evidence                                                                                                                                                                   |
|----------------------|-----------------------------------|--------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exact                | Full history                      |        | all canonical records                                | control and repeatability baseline                                                                                                                                                 |
| Conservative         | wide Recent Frontier              |        | prompts, recent mutation/verification/protocol spine | current bounds session growth; parameters require workload gate                                                                                                                    |
| Balanced candidate   | repeat-qualified Frontier (M2/P1) | Recent | same floors with bounded older causal groups         | matched N=5 preserves 3/3 Full successes at 37.51% clean paired saving (29.05% failure-aware); selective N=6 resolves 5/6 at 51.36% own saving but lacks a Full Task-6 denominator |
| Experimental economy | Prompt Pinned (E2+F1C)            |        | all prompts, active/recent epochs, closure evidence  | 3/3 cold identities; hybrid receipts prevent strict native promotion                                                                                                               |

Table 5: Evidence-based profile interpretation. Names express operating intent; they do not convert discovery cohorts into population guarantees.

Promotion requires a larger held-out task cohort, repeat runs clustered by task, another model family, and another agent harness. The same selector must also be tested in both one-task-per-session and persistent multi-task modes. A policy that helps only after explicit task segmentation is not a general session-memory policy; a policy that retains all prior independent issues cannot deliver the asymptotic benefit that motivates native addressable K/V.

The practical default-selection rule is correspondingly simple: prefer the least aggressive profile that meets the application’s memory bound, and fall back toward Full when dependency evidence, protocol state, or model behavior is uncertain. Savings are accepted only when official resolution is preserved on the paired eligible cohort and aggregate calls do not increase. This rule turns the paper’s negative results into an operational guardrail rather than a claim that all semantic selection is unsafe.

## 7 Frozen Runtime Handoff

Paper 8.5 freezes record identities, spans, original positions, replacements, and a plan digest. It does not infer engine performance from logical tokens. For a selection-only plan, the cumulative selected original-token count is the engine-independent upper bound on reusable K/V demand. Paper 4.5 must replay the identical ledger and report page rounding, resident references, physical copies, newly evaluated suffix tokens, and temporary bytes separately. A logical saving difference across engines is a failed ledger comparison, not an engine result.

Earlier runtime-transfer diagnostics preceded the current engine fixes and are not used as post-fix evidence here. They also mixed strict selection with model-visible closure receipts. A receipt created after source K/V exists is transformed context: its tokens require evaluation and cannot be credited as native reuse. The post-fix Paper 4.5 handoff will therefore map only frozen selection-only plans directly to K/V omission; ingest-time compaction and post-cache transformation remain separate treatments.

## 8 Discussion and Limitations

**What the positive points establish.** Recent Frontier and Prompt Pinned are two independently motivated mechanisms that cross the intended 30–50% logical-saving band on small qualified cohorts while preserving their paired successes. They show that useful agent history is not synonymous with a blind suffix and that substantial removable history can exist without an explicit task-boundary signal. They do not yet estimate population accuracy, establish a production default, or show that the same parameters transfer unchanged across agents and models.

**What the negative points establish.** Whole-record semantic deletion fails for a deeper reason than an unfortunate threshold: current-world authority and reasoning utility have different lifetimes. Likewise, a low per-request retention ratio can be economically negative when it causes recovery. These observations constrain future policy design: use negative selection conservatively, retain protocol and mutation floors, and promote policies on paired cumulative work rather than on nominal token omission.

**Closed-task conditioning is neither uniformly helpful nor safely recoverable from arbitrary old K/V.** The frozen Task 4/Task 7 diagnosis sharpens that constraint. Removing all closed-task exemplars repairs stale-task continuation on Task 4 (12 calls versus 13 for FULL, with 68.01% paired token saving), but Task 7 then fails at the 40-call limit although every active-task message is retained. Its own-trajectory saving is 80.70%, yet the longer trajectory reduces raw paired saving to 10.65% and failure-aware saving to zero. Old histories can therefore provide useful workflow conditioning while also supplying harmful task identity. Selecting only old command-token K/V is not a task-neutral repair: those vectors were themselves computed while attending to the old task and rationale. A short task-neutral workflow anchor encoded once in the static system prefix repairs the frozen first-action failure on both Task 4 and Task 7: P0/W0 reduces the reconstructed requests by 93.1% and 93.0%, respectively, with stable bounded searches in 2/2 repeats per arm.

The subsequent autonomous Task 7 pair exposes two further controls. First, nominal temperature zero is not operationally deterministic on this endpoint: the served Modelfile retains temperature 0.7, top- $k$  20, and top- $p$  0.8 defaults, and exact replays vary from one valid action to twelve code blocks. Declaring top- $k = 1$  makes the selective first request identical in 5/5 repeats and is now a mandatory pairing coordinate. Second, under this genuinely greedy control, anchored FULL and P0/W0 both resolve officially. P0/W0 selects 52,807 cumulative input tokens versus Full’s 213,287, for 75.24% clean paired saving (89.84% against its own 519,679-token counterfactual), but uses 14 calls rather than six. The extra work is an inefficient search followed by fragile Bash `sed` insertion and repeated inspect/revert/re-edit cycles, even though every active-task message is kept. Keeping the immediately preceding task whole restores the targeted first search but repeats the same edit loop and is stopped after 15 recorded selections. Thus capability and input economy pass on this identity, while the action-cost gate fails. The next treatment must provide a task-neutral pre-ingest

reliable-edit affordance (or a typed edit primitive), identically to Full and selective arms; it must not summarize or re-encode resident K/V.

Follow-up mechanism tests separate localization, mutation, and termination. A prose guarded-edit anchor worsens matched Full from six to 11 calls and its selective arm is stopped after 18; an old mutation-verification exemplar leaks pytest workspace identity into Django. Frozen replay instead selects a transactional edit in 3/3 trials, while an explicit search example emits multiple actions in 3/3 and is rejected.

Installing the digest-recorded transactional command exposes and fixes a receipt bug that misclassifies replacement text `self.dict` as a second file. With that parser fixed, the post-fix arms share the anchor, tool, generation coordinates, six-episode prefix, and 1,024-token completion ceiling. Full resolves in six calls/six actions using 213,298 tokens; P0/W0 resolves in eight/seven using 30,647 selected tokens from a 297,431-token counterfactual: 89.70% own and 85.63% paired saving, with no reacquisition. One additional call is rejected before execution for length and the action delta is one focused verification. A 2,048-token ceiling instead reaches 20 calls/19 actions and fails officially. Thus capability and saving pass on this identity, but the no-call-increase gate does not; the `v8_transactional_matched_task7` bundle is not a production profile.

A prospective Task 4 admission binds v7 to the current instruction. Frozen pre-edit replay is repeatable in 3/3 trials, yet autonomous FULL reaches 25 calls (eight searches, twelve reads, no test), never invokes the transactional command, and does not submit; no selective arm is admitted. Prompt placement is therefore insufficient, and typed-action reliability remains a separate gate. See `v9_task4_current_instruction_anchor`.

**Oracle scope.** Leave-one-bundle-out and bounded subset search answer whether headroom exists on a frozen successful trajectory. They cannot predict the new trajectory created by omission, and independent removability does not imply joint removability. Oracle results are therefore hypothesis generators and add-back diagnostics, not deployable policies or accuracy estimates.

**External validity.** The autonomous evidence uses mini-swe-agent, a Bash tool surface, a small number of SWE-bench Verified identities, and primarily Qwen3-Coder-30B. Bash is useful because file reads, writes, searches, tests, and return codes can be typed from commands and observations. Controlled Pi, Kilo, and PRA-Agent admissions now establish that two native typed-tool protocols and one record-native agent all solve the same locked task under exact FULL controls. These are compatibility results, not policy-accuracy estimates: future work must still reproduce the savings findings across the locked cohort, then cover a current richer open-agent runtime, another model family, longer dependent sessions, and task-clustered confidence intervals.

**Runtime boundary.** Logical omission says which existing records a model should see. It does not say whether an engine re-encodes text, copies K/V, preserves original positions, or creates large temporaries. Post-cache receipts also introduce new tokens and must be charged. Paper 4.5 receives the frozen plan and owns those physical and numerical gates.

## 9 Conclusion

Agent history is neither a bag of tokens nor merely current world state: provenance, control, and semantic evidence can outlive the state they create. Boundary-free policies expose removable history, but current cohorts remain small. The defensible endpoint is official resolution, calls, and cumulative work—not nominal retention. Logical policy and physical realization remain separate contracts.